

AutoEncoder on MNIST - Embedding Analysis Report

- **Introduction**

This report details using a Convolutional AutoEncoder on the MNIST dataset, encoding images into lower-dimensional spaces for reconstruction. Various embedding sizes (2, 5, 10, 20) were tested. PCA and t-SNE visualized the embeddings, with GIFs showing latent dimension manipulations.

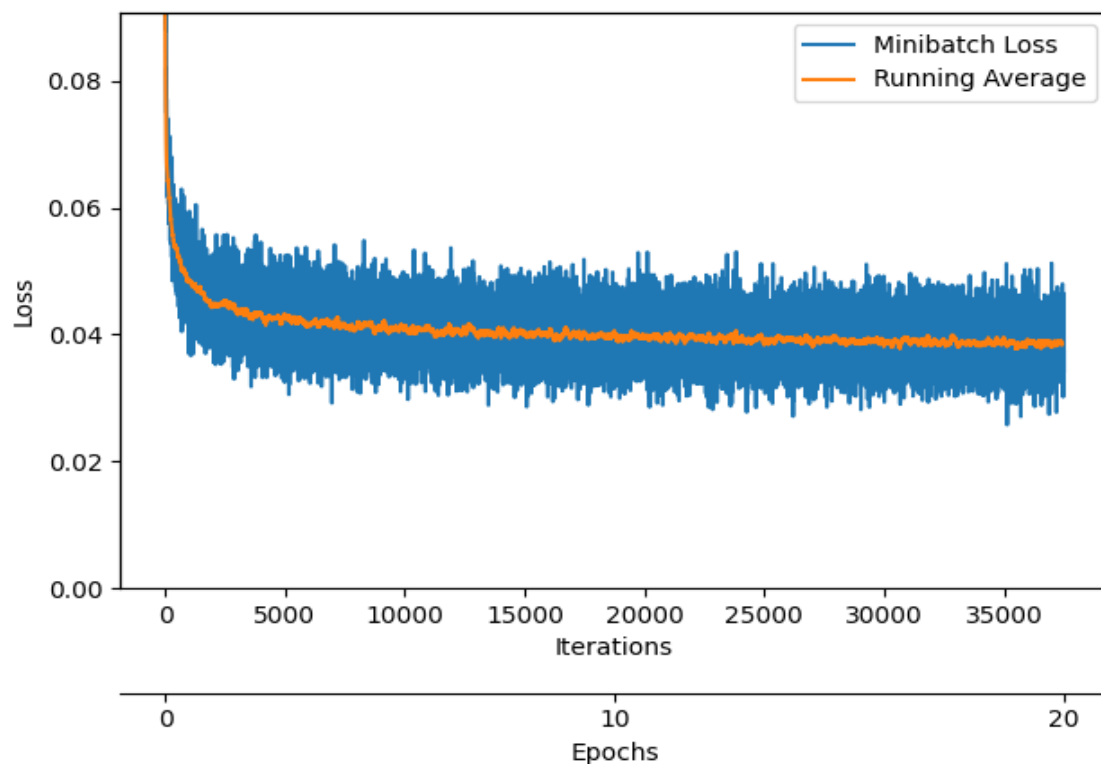
- **Tools and Techniques**

- Dataset: MNIST (28x28 grayscale images of handwritten digits).
- Model: Convolutional AutoEncoder implemented in PyTorch
- Embedding Sizes: 2, 5, 10, 20.
- Visualization: PCA and t-SNE for latent space projection.
- Output: Loss plots, reconstructed images, latent space scatter plots, embedding histograms, and GIFs

- **Embedding Size = 2**

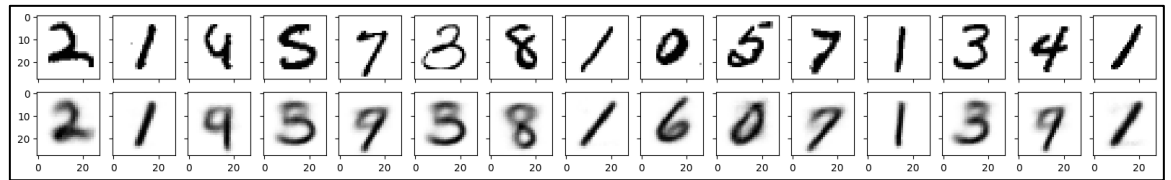
- **Training Loss**

Loss curve showing minibatch loss and running average.

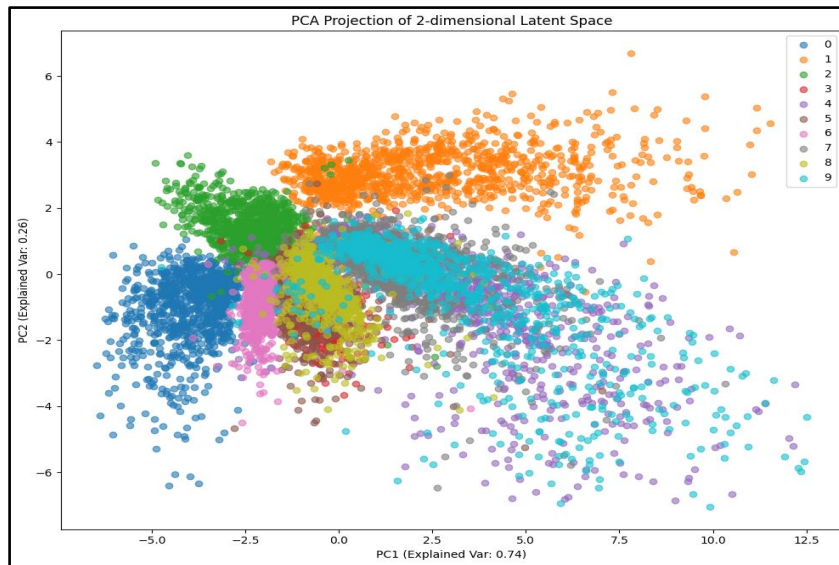


B

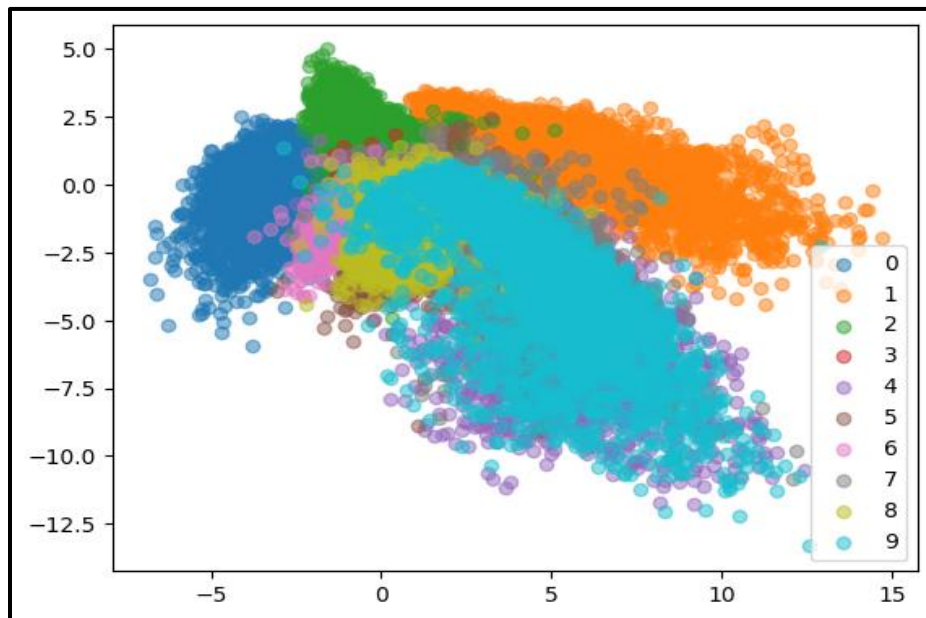
- Reconstructed Images



- latent space of the generated embeddings



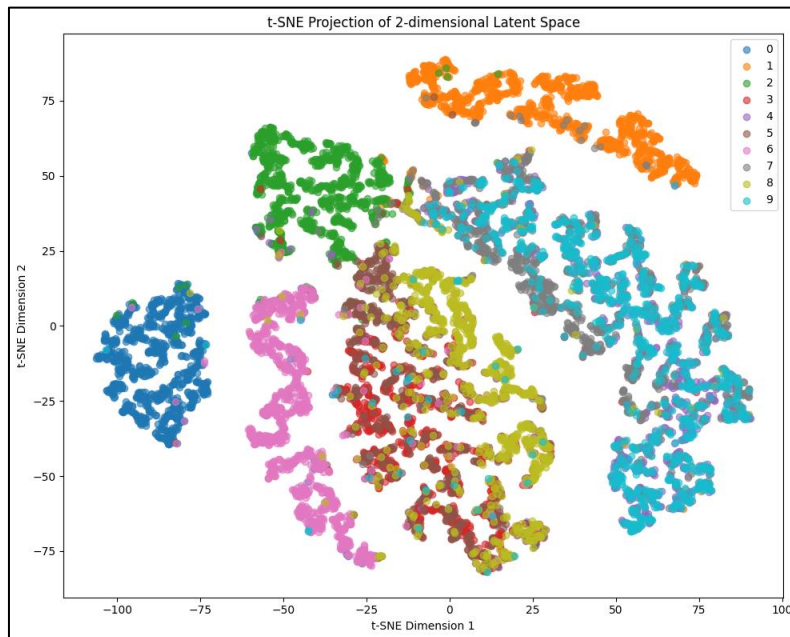
- Latent Space Visualization (PCA)



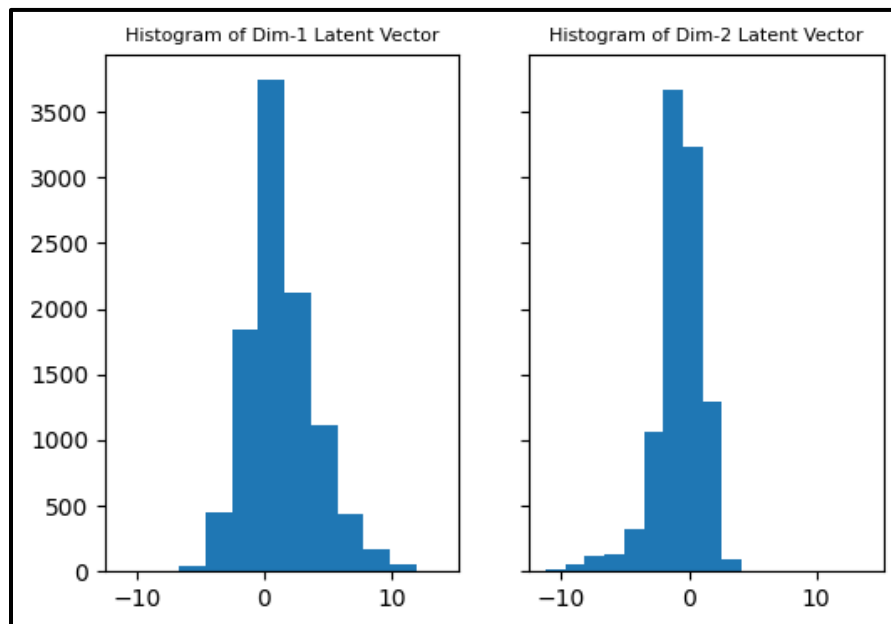
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- **Latent Space Visualization (t-SNE)**

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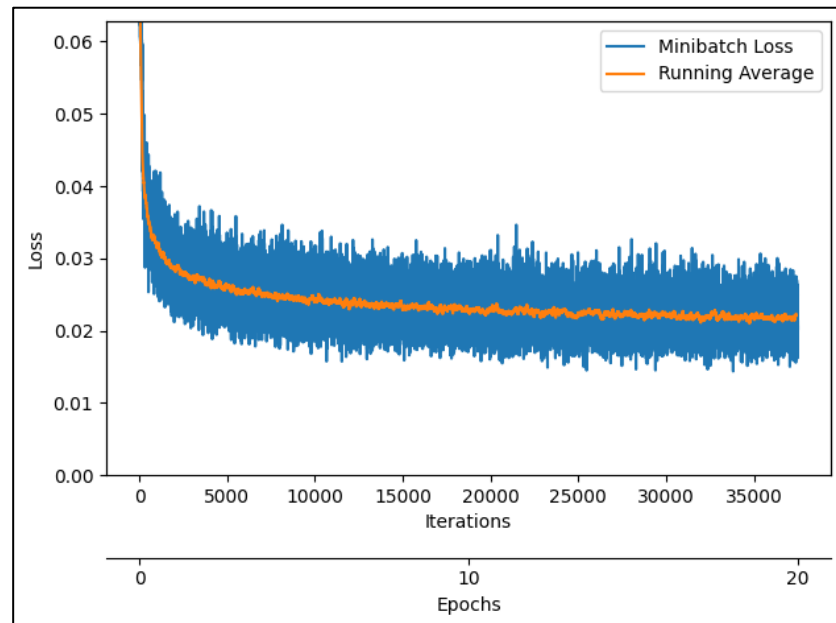
- **Embedding Histograms**



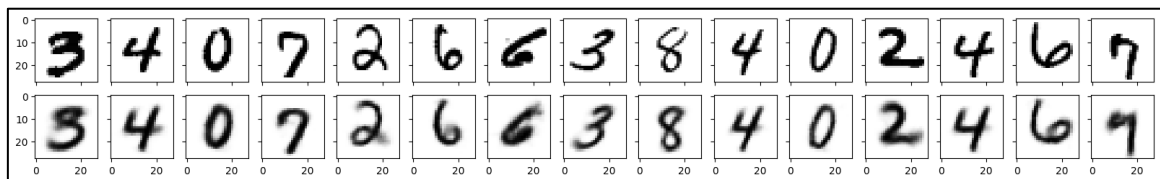
Embedding Size = 5

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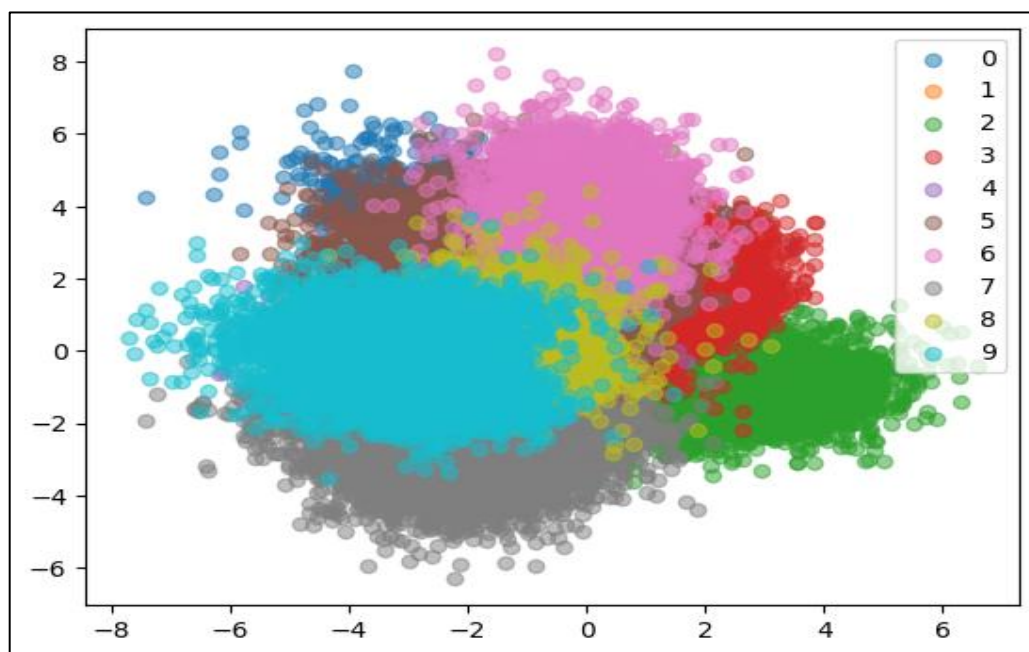
- **Training Loss**



- **Reconstructed Images**

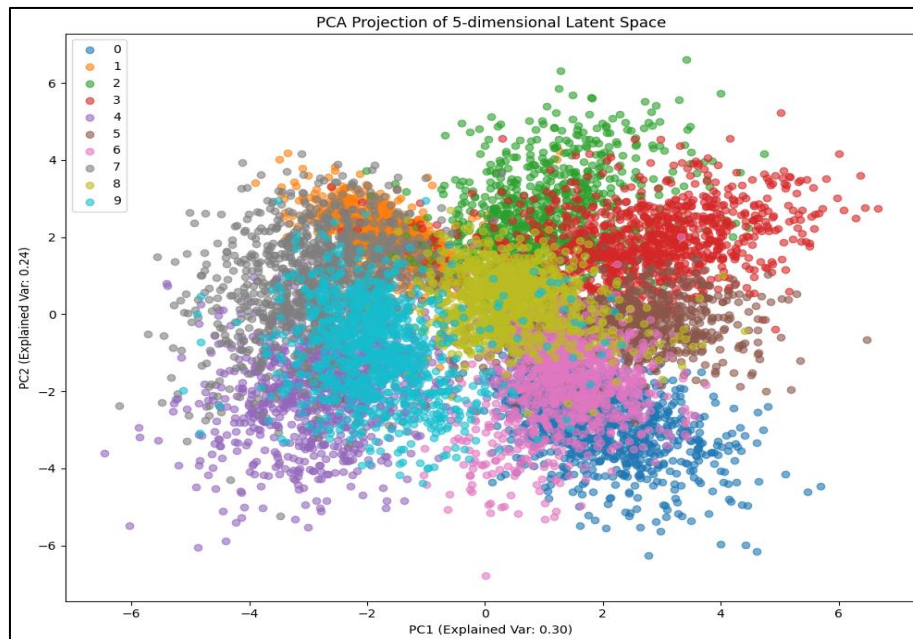


- **latent space of the generated embedding**

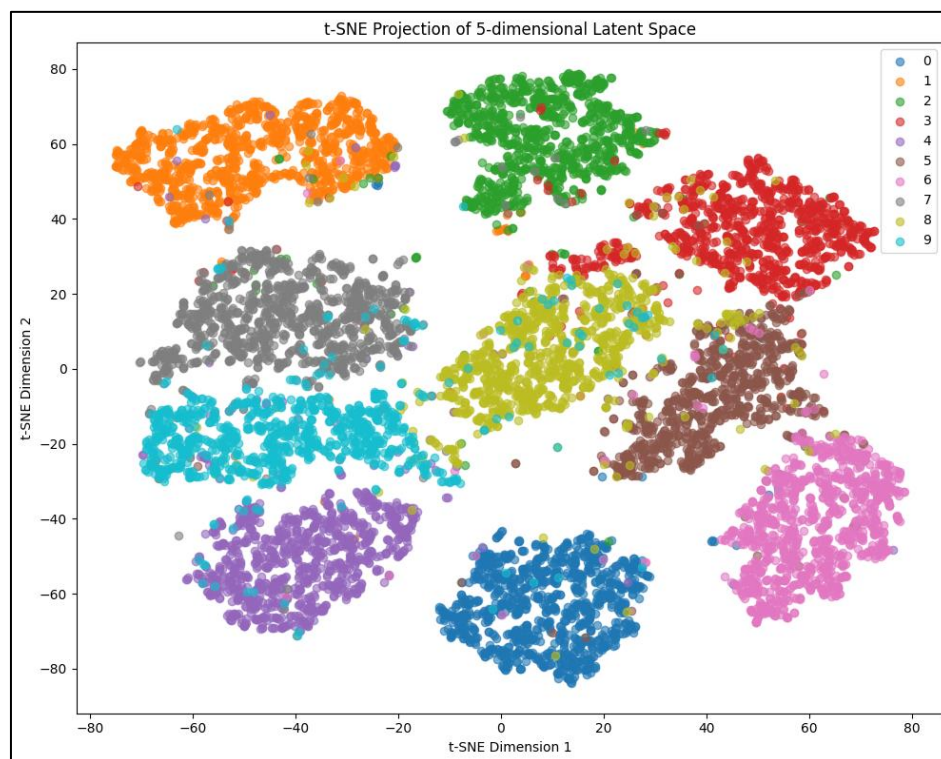


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- Latent Space Visualization (PCA)

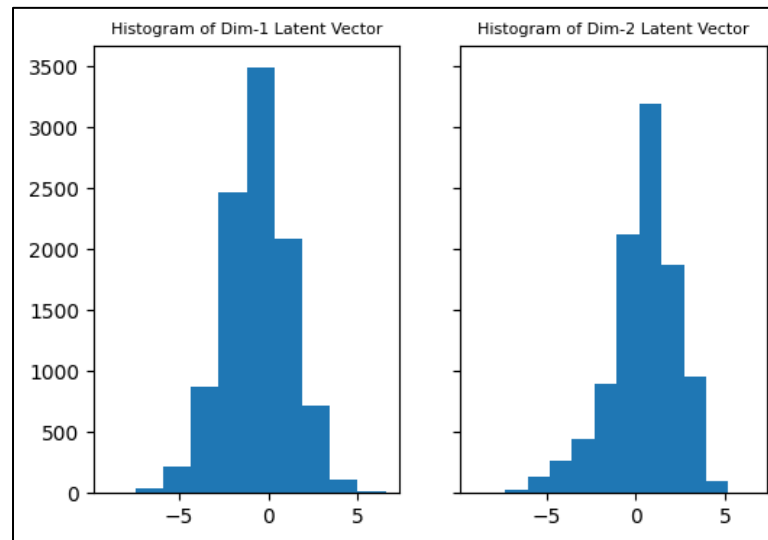


- Latent Space Visualization (t-SNE)



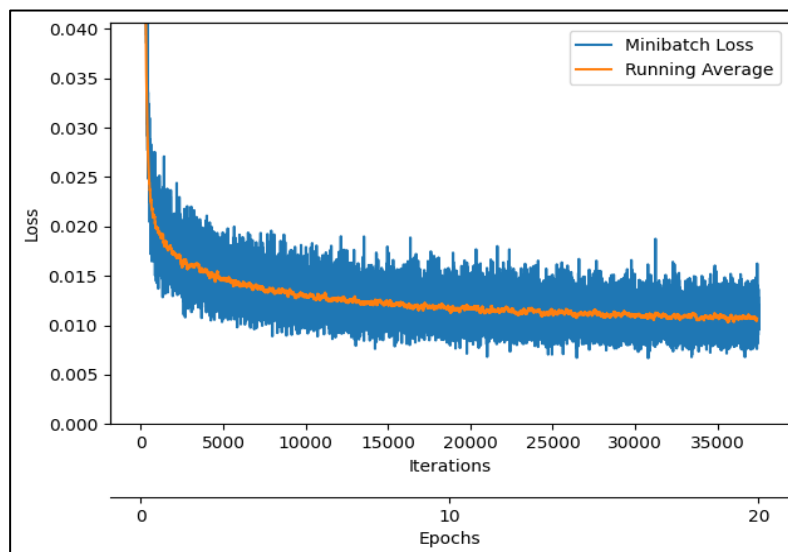
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- **Embedding Histograms**

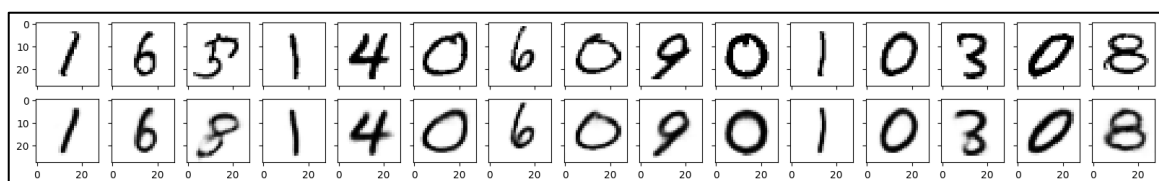


Embedding Size = 10

- **Training Loss**

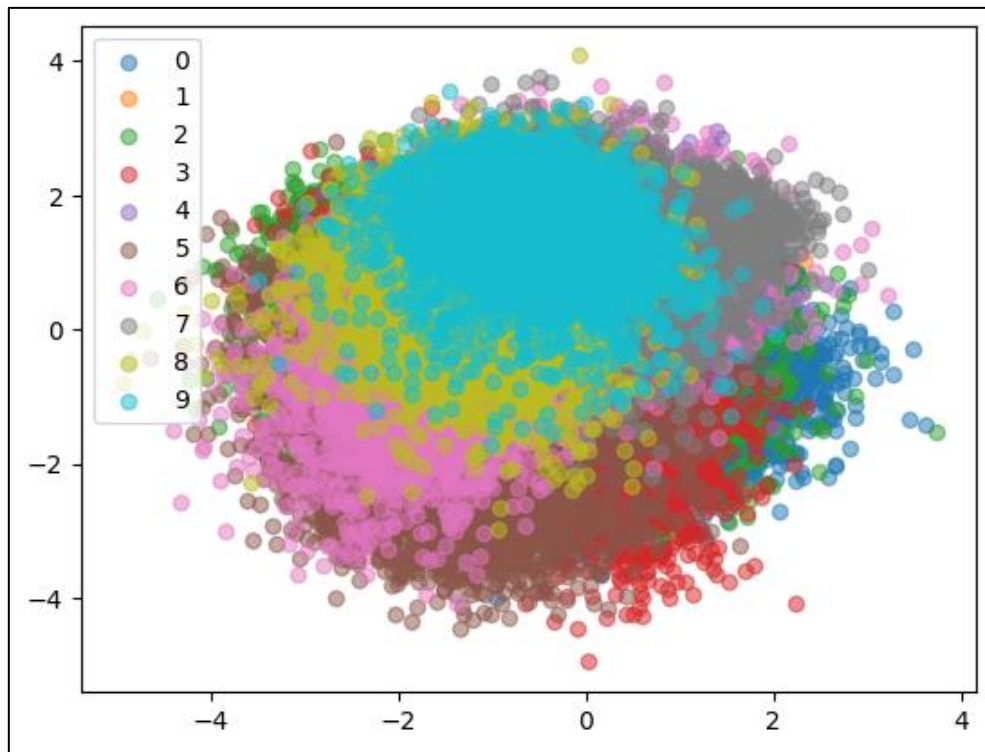


- **Reconstructed Images**

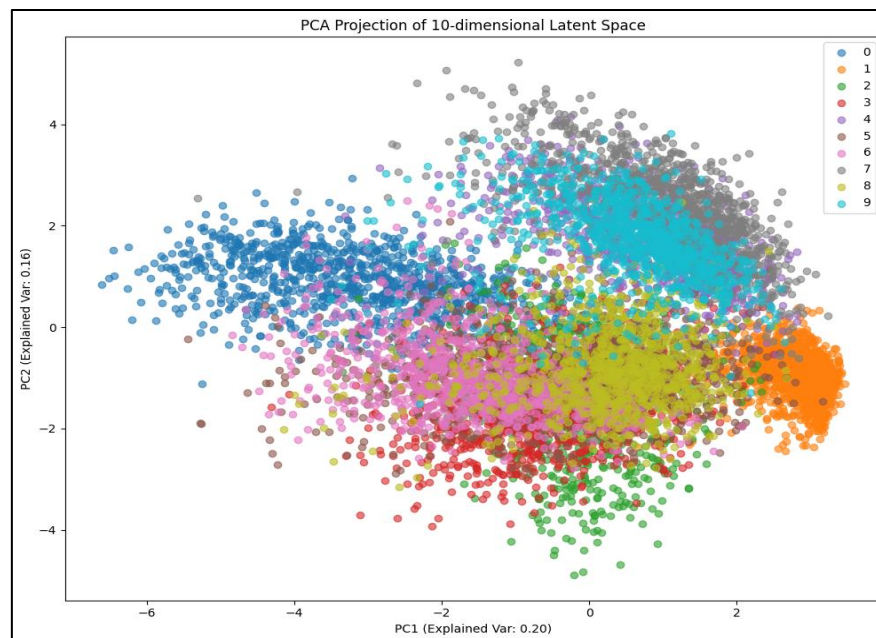


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- latent space of the generated embedding

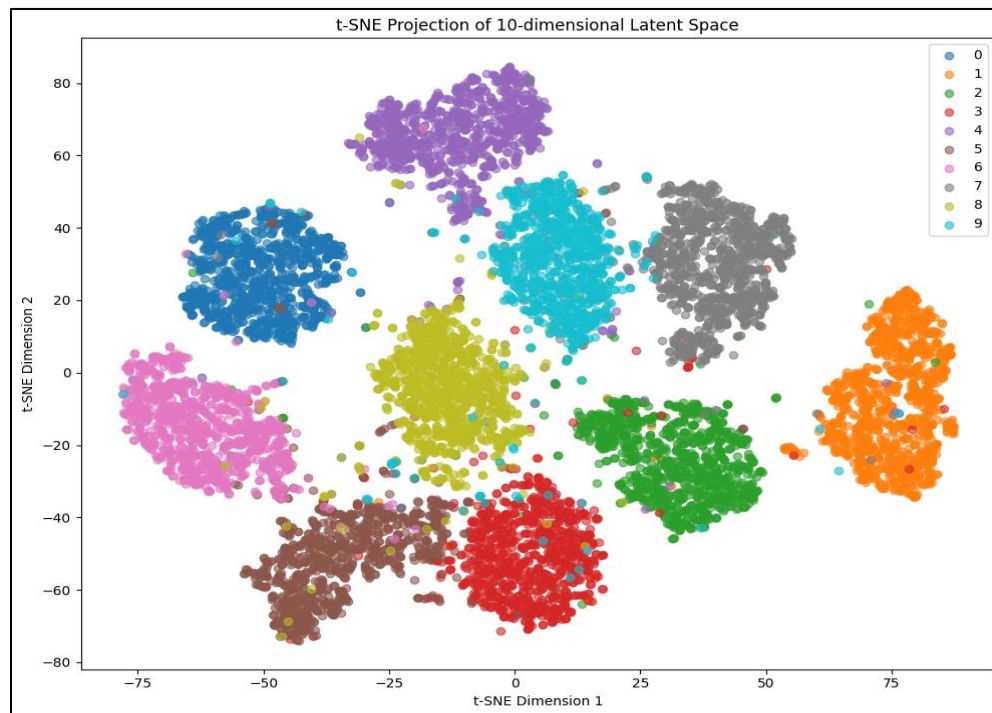


- Latent Space Visualization (PCA)

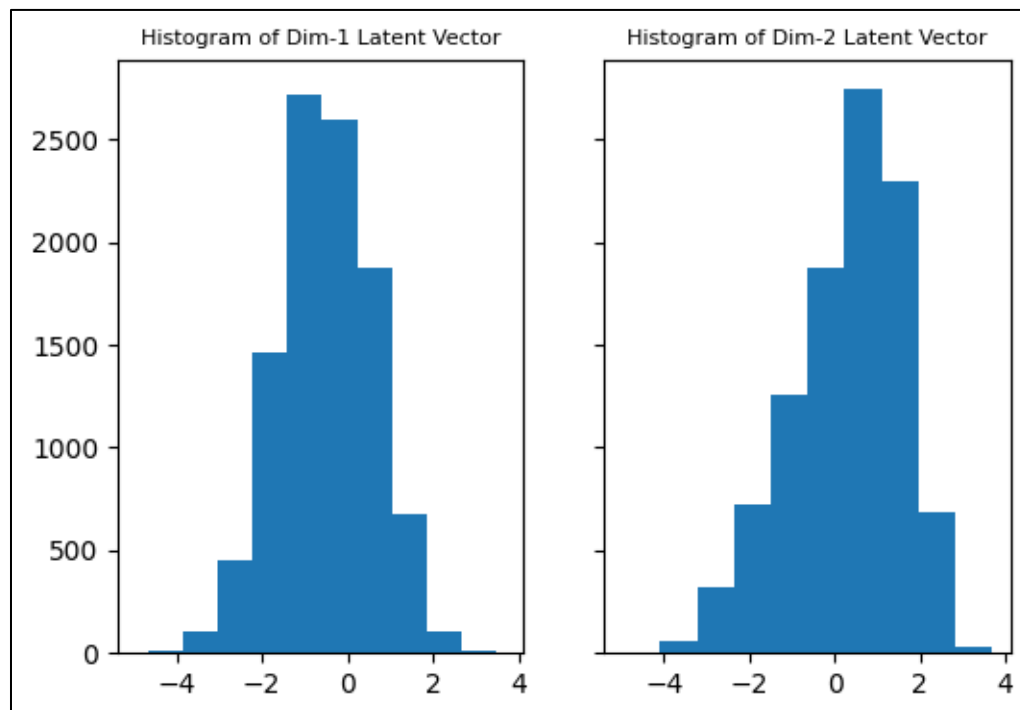


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- **Latent Space Visualization (t-SNE)**



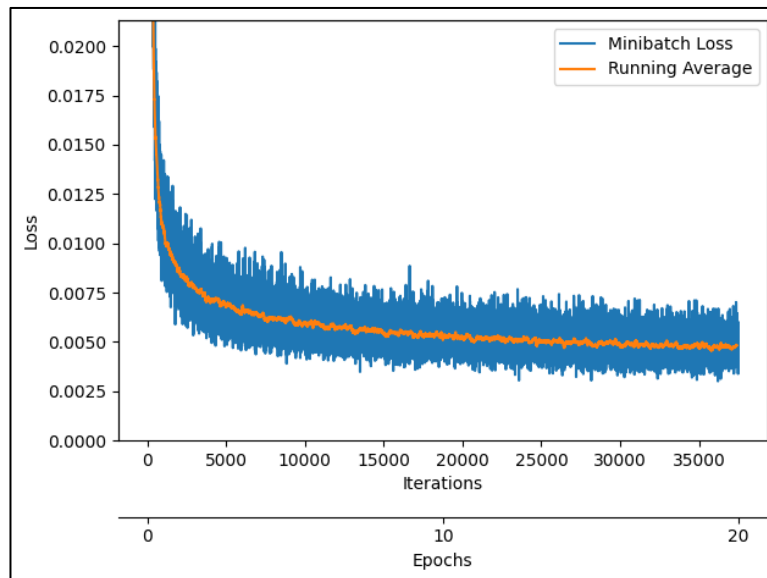
- **Embedding Histograms**



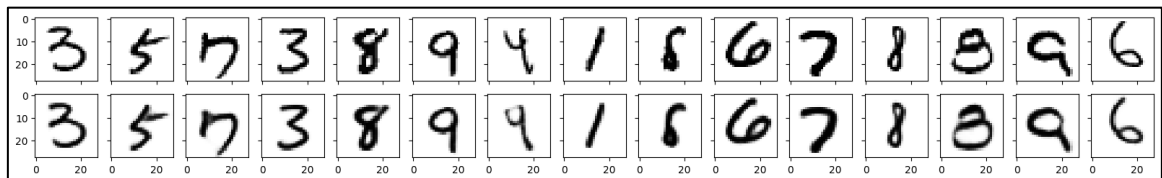
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Embedding Size = 20

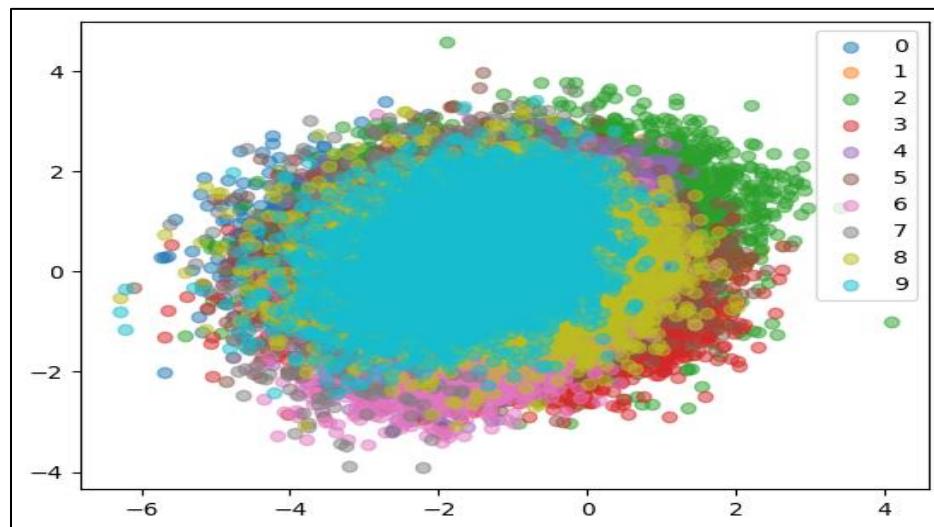
- Training Loss



- Reconstructed Images

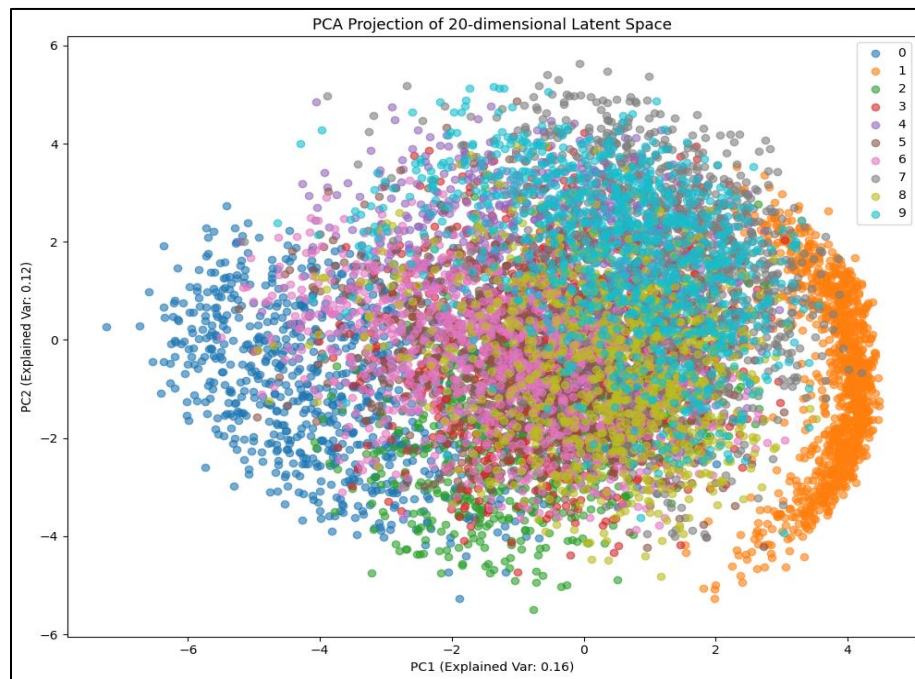


- latent space of the generated embedding

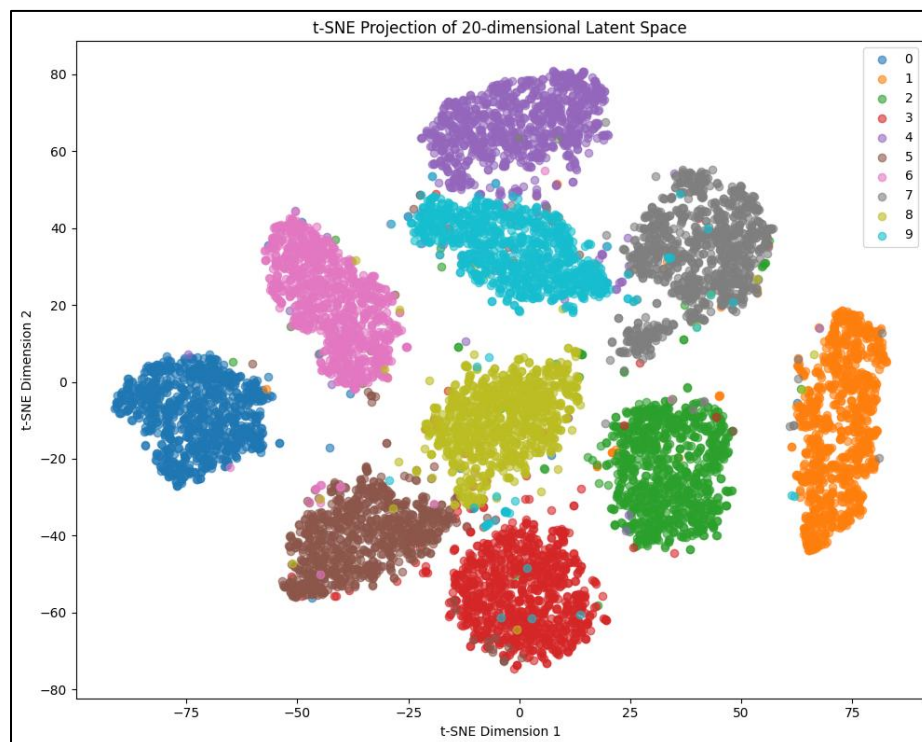


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■ Latent Space Visualization (PCA)

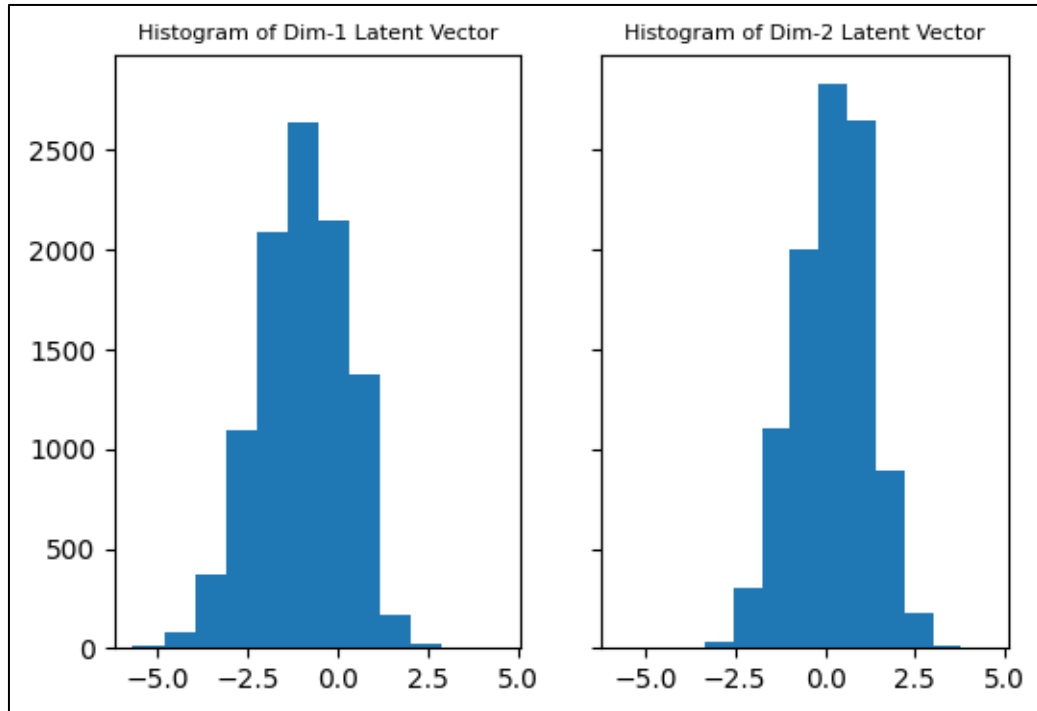


■ Latent Space Visualization (t-SNE)



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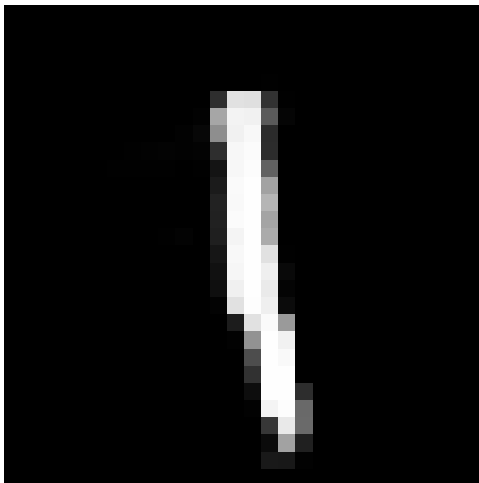
- **Embedding Histograms**



Latent Dimension Interpolation (Embedding Size = 2)

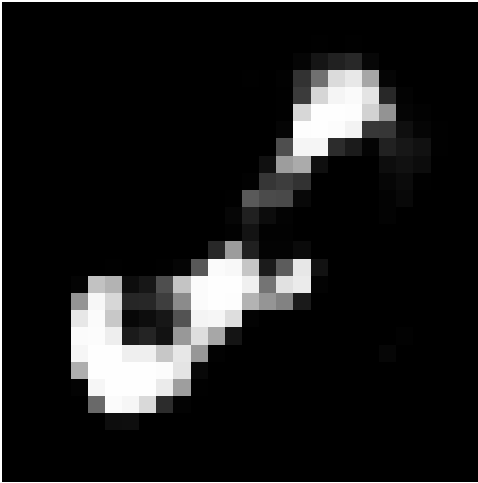
These animations show the reconstructed image as we vary one latent dimension at a time.

GIF - Varying Dimension 1



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GIF - Varying Dimension 2



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